Advanced Data Analytics

Excel Project

❖ Data Analytics

- Process of exploring and analysing data to gain insights and drive informed decision making within organizations.
- Encompasses a wide range of techniques, methodologies, and tools aimed at leveraging data to improve business performance, enhance operational efficiency, and achieve strategic objectives.

❖ Different Stages in Data Analytics

- Collection of relevant data from various sources, including internal databases, external sources, and third-party data providers.
- Clean, integrate, and transform the data into a format suitable for analysis.
- Visually explore and summarize data to identify patterns, trends, and relationships.
- Apply statistical methods and models to quantify relationships between variables, test hypotheses, and make predictions.
- Forecast future outcomes or trends based on historical data and statistical algorithms

❖ Application of Business Analytics

- Customer Segmentation
- Sales Forecasting
- Supply chain optimization
- Risk Analysis
- Operational Efficiency
- Predictive Maintenance

Problem 1: **Prescriptive Model Pricing**

- A firm wishes to determine the best pricing for one of its products in order to maximize profit.
- Analysts determined the following model: Sales = -2.9(price) + 3000
- Total revenue = (price)(sales)
- Fixed cost = 5000
- Total Cost = 10(Sales) + 5000
- Identify the price that maximizes profit, subject to any constraints that might exist.

Sales = -2.9 (Price) + 3000
Total revenue = (Price)(Sales)
Cost = 10(Sales) + 5000

Price	**$ 100**
Sales	2950
Total Cost	$ 34,500
Cost per unit	$ 12
Profit	$ 260,500

Problem 2:**Sales Promotion Decision**

- In the grocery industry, managers typically need to know how best to use pricing, coupons and advertising strategies to influence sales.
- Controlled experiments : Grocers implement different combinations of pricing, coupons, and advertising to observe the resulting sales. Analytics helps develop a predictive model of sales as a function of these decision strategies.

❖ Sales Prediction with Multiple Regression Analysis

- Find the pricing and Ads Budget to maximize the Sales
- Use Multiple Linear Regression
- Cost price of the Product is $8
- Minimum Profit desired is $10,000

Cost Price per unit	$9.00
Minimum Profit	$4,000.00

	Coefficients
Intercept	3092.883488
Price	-142.5484493
Ads Budget	0.10225716

Maximum Ads Budget	$10,000.00
Minimum Ads Budget	$1,000.00

Results from Regression Analysis:
Solution: Sales = -142 x Price + 0.10 x Ad+3091

	Sales Promotion Dat a		
Week	Price	Ads Budget (Thousands USD)	Sales
1	$15.0	1500	1240
2	$16.4	4000	950
3	$13.0	2000	1580
4	$16.5	3000	1190
5	$12.0	1000	1430
6	$11.0	1000	1560
7	$14.3	1000	1060
8	$18.2	3465	800
9	$9.6	1000	1895
10	$15.4	4000	1475
11	$12.0	2000	1560
12	$14.0	1000	1235
13	$16.0	3000	1065
14	$15.0	1000	1025
15	$11.0	2000	1645

Problem 3:**KERC INC – Car Manufacturer**

Has two main car models

- KERC 501
- KERC 601

Below table provides the Activity times (hrs) of the models

	Activity time in Hours		
Car Model	Assembly (All parts)	Frame Fitting and Painting	Quality Assurance
KERC 501	8	4	2
KERC 601	10	3	1.6

Car Model	Profit (USD Thousands)
KERK 501	4.9
KERK 601	5.8

Manufacturing Activity	Time Available Hours per Week
Assembly (All parts)	10800
Frame Fitting and Painting	4500
Quality Assurance	2500

❖ Optimize KERC production

Define Objective Function (profit) and find the Optimal manufacturing quantities for car models to maximize the objective function?

Problem 4:WonderZon Retailer

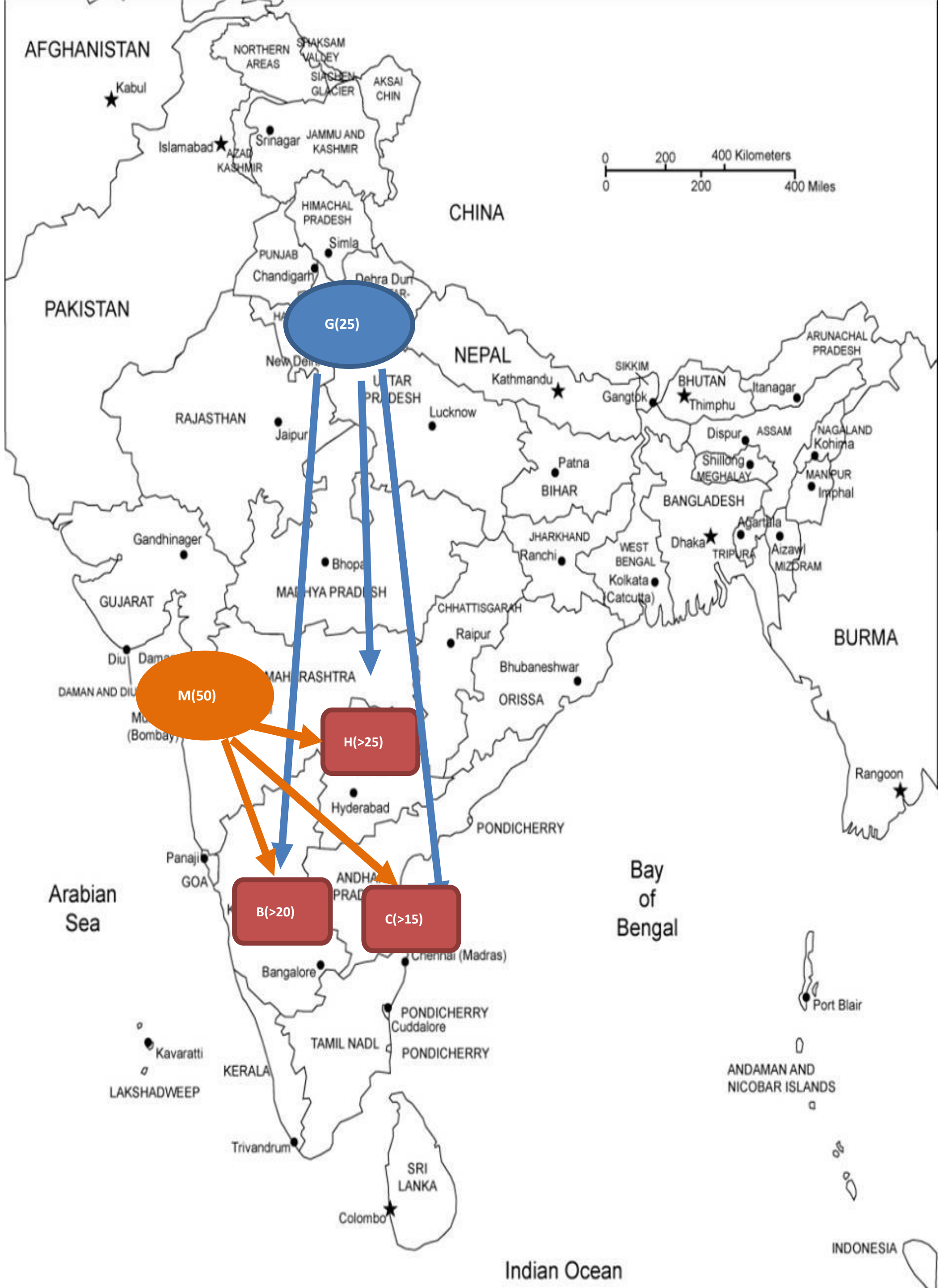
- Optimization Model of Logistics to minimize cost
- Predictive Analytics with Less uncertainty

Warehouses	Shipment Quantity
Gurgaon	25
Mumbai	50

Delivery Centers	Minimum Quantity
Bengaluru (B)	20
Hyderabad(H)	25
Chennai(C)	15

Logistics Cost in INR per Package			Delivery Centers	
	From / To	Bengaluru (B)	Hyderabad (H)	Chennai (C)
Warehouses	Gurgaon (G)	116	106	115
	Mumbai (M)	110	95	112

- Find Objective function for cost and optimize the network for min logistics cost for WonderZon Retailer



Problem 5:Internet Usage - Plan Decision

- Predictive Analytics
- with High Uncertainty
- Monte Carlo Simulation

SMART IT Inc. Internet Plan

Corporate Internet Plans	
FLEXI	Flat plan charging $2 per GB
FAST1000	1000 GB included , Post 1000 GB , $1500 + $3.5 per GB

- using FLEXI corporate internet plan for a year and thinking about deciding to switch to FAST1000. SMART IT INC business is seasonal and internet usage varies as per the demand.
- Generally bill usage vary from around 1000 GB. Past 6 months usage in GB are 620,1050,1200,1480,870,1600
- Find the best plan for SMART IT (Use Monte Carlo Simulation with Normally distribution data)

Problem 6:**Kickathlon Sport Retailer**

- Decision Tree & Monte Carlo Simulation
- It is a probability model used to predict the probability of various outcomes.
- It is used to mitigate the possibility of unpredicted outcomes.
- Predictive Analytics with High uncertainty.

Kickathlon – Supplier Decision Making

- Kickathlon, a French based Sports Retailer, plans to introduce a new product, TrekBareFoot – a tough Polyester based Socks, which can protect the foot while trekking barefoot.

	USA (A)	China (C)
Supplier Capacity	6000	15000
Fixed Contract Charges	0	$500,000
Unit Product cost	$150	$130
Profit per unit	$50	$70

❖ Kickathlon Sport Retailer

- TrekBareFoot is scheduled to launch in next Summer beginning so should be manufactured well in advance.
- The price tag for TrekBareFoot is fixed at **$200/-**
- If the marketing efforts create a strong demand then about **15000** units can be sold, on the lower side, weak demand might end up with only **6000** units.
- There is a 50:50% chance for strong and weak demand in the market.
- TrekBareFoot being a specialized product, Kickathlon found only two suppliers one in USA and other in China. The pricing is as below.

❖ Decision Trees

- A Decision Tree is a visual and analytical decision support tool that helps map out decisions and their possible consequences, including risks, costs and benefits.

Decision Tree – Key Components

- Decision Nodes

There are decision points where it branches to more than one decisions

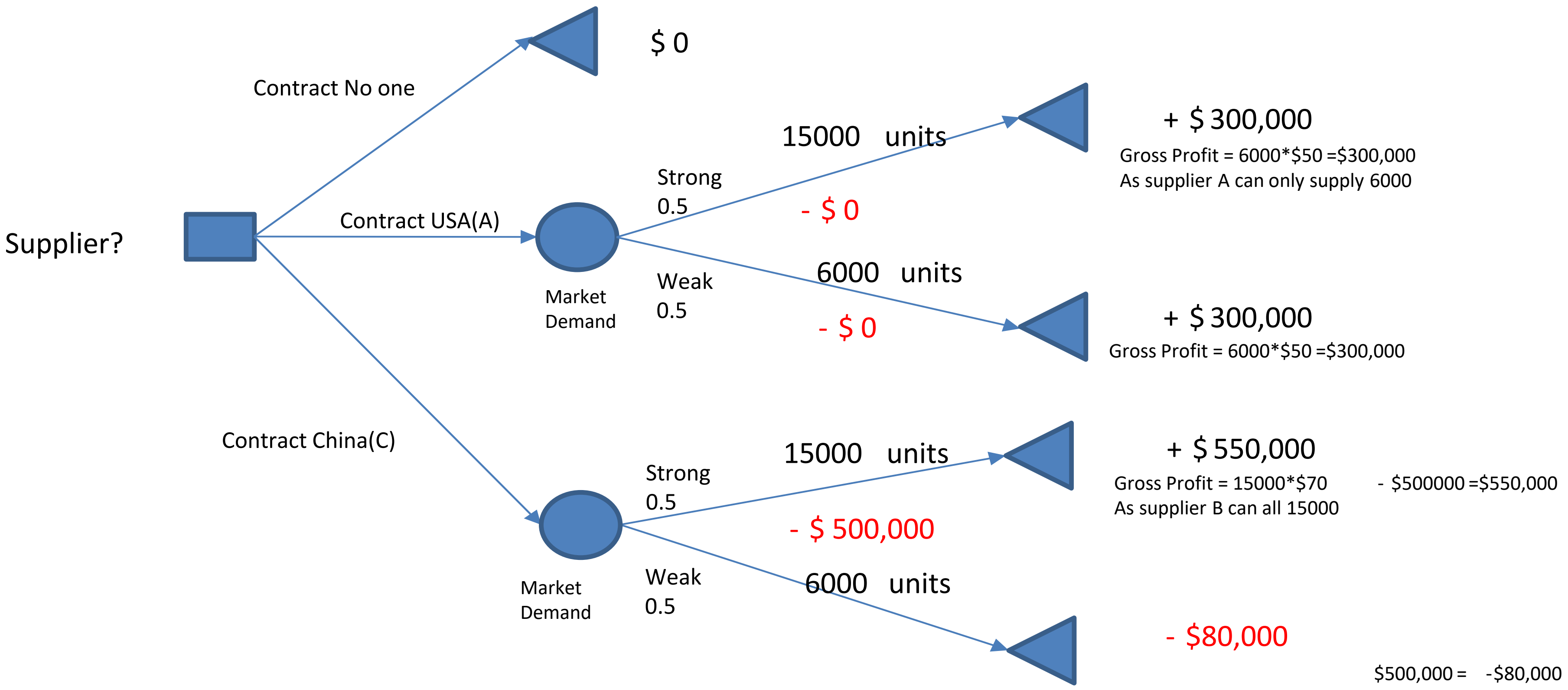
- Chance / Event Nodes

There are points of uncertainty, which is defined by probability / random variable

- Outcome Nodes

There are the outcome variable (Objective function), which needs to be either maximized or minimized

❖ Kickathlon – Supplier Decision Tree



❖ What-If Analysis

- **What-If Analysis** is a decision-making tool that helps you understand the **impact of changing input values** on the **outcome** of a model or formula
- In simple terms:
- "What will happen to the result **if** I change this input?"

Types of What-If Analysis

1. Scenario Manager
Allows you to create and compare multiple input scenarios.
Example: Best Case, Worst Case, Most Likely Case.
2. Goal Seek
Works **backward**: You know the result you want and want to find the input needed.
Example: What interest rate is needed to achieve ₹10,000 return?
3. Data Table
Shows how changing one or two inputs affects the result.
Types: One-Variable and Two-Variable Data Tables.

Basic Excel Functions

- Remove Duplicate
- Splitting Text
- Trim Space
- Change Case
- Conditional Formatting

Advanced Excel Functions

- Vlookup
- Hlookup
- Xlookup
- Index and match
- Sumif and sumifs
- Countif and countifs
- Unique